Grade 10 – Term 1
Decision Trees with Staged Filtering: C9 Practice

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Evaluated criteria

C9 Constructs and interprets decision trees for staged filtering rules, applying decision criteria in the specified order to identify the final alternative.

1 Introduction

Use the already calculated EV, Maximin, and Maximum Regret summary tables directly; do not recompute these statistics from the payoffs. Apply the filtering rules in the specified order. At each stage, compare only the alternatives retained by the previous stage. Higher EV and Maximin values are preferred; smaller maximum regrets are preferred. Keep the supplied regret values fixed while filtering.

Represent each sequential decision process in a four-column filtering table: # Step, Description, Left Alternatives, and Eliminated Alternatives. Start with all alternatives at step 0; then record the criterion, filtering rule, relevant values, and alternatives retained or eliminated at steps 1–3. Identify the final selected alternative in the last row. Retain tied alternatives together; do not introduce an arbitrary tie-breaking rule. State probabilities belong to the payoff model used to calculate EV.



2 Decision Trees with Staged Filtering

Problems:

2.1. City logistics policy

2.2. Regional distributor expansion

2.3. Trade finance guarantees

2.4. Insurance portfolio mixes

2.1 City Logistics Policy

Problem Description

A city logistics office must choose one policy among $B_1, B_2, B_3, B_4, B_5, B_6$. Payoffs (thousand USD) and probabilities are:

State	Probability	B_1	B_2	B_3	B_4	B_5	B_6
T_1	0.50	95	90	110	85	100	92
T_2	0.30	70	75	60	80	65	78
T_3	0.20	30	40	10	55	35	20

The summary statistics below are already calculated. Use them directly; do not recompute them from the payoffs.

Alternative	EV	Maximin	Maximum Regret
B_1	74.50	30	25
B_2	75.50	40	20
B_3	75.00	10	45
B_4	77.50	55	25
B_5	76.50	35	20
B_6	73.40	20	35

Filtering rule: (1) keep the best 4 EV options; (2) from those, keep the best 2 Maximin options; (3) from those, choose the smallest Minimax Regret.

Questions/Tasks



C9 Apply the filtering rules using the supplied results. Construct a step-by-step filtering table showing the rule, relevant numerical values, and alternatives retained and eliminated at each stage, including the final decision.

Solution

# Step	Description	Left Alternatives	Eliminated Alternatives
0	Initial alternatives	$\{B_1, B_2, B_3, B_4, B_5, B_6\}$	—
1	EV: keep the 4 highest values.	$B_4(77.50)$, $B_5(76.50)$, $B_2(75.50)$, $B_3(75.00)$	$B_1(74.50)$, $B_6(73.40)$
2	Maximin: keep the 2 highest remaining values.	$B_4(55)$, $B_2(40)$	$B_5(35)$, $B_3(10)$
3	Minimax Regret: select the smallest remaining maximum regret.	**Selected:** $B_2(20)$	$B_4(25)$



2.2 Regional Distributor Expansion

Problem Description

A regional distributor must choose one expansion plan among $X_1, X_2, X_3, X_4, X_5, X_6$. Payoffs (thousand USD) and probabilities are:

State	Probability	X_1	X_2	X_3	X_4	X_5	X_6
U_1	0.30	130	120	140	110	125	118
U_2	0.25	90	95	80	100	88	92
U_3	0.25	60	70	55	85	65	75
U_4	0.20	20	40	10	60	45	35

The summary statistics below are already calculated. Use them directly; do not recompute them from the payoffs.

Alternative	EV	Maximin	Maximum Regret
X_1	80.50	20	40
X_2	85.25	40	20
X_3	77.75	10	50
X_4	91.25	60	30
X_5	84.75	45	20
X_6	84.15	35	25

Filtering rule: (1) keep the best 3 EV options; (2) from those, keep the best 2 Minimax Regret options; (3) from those, choose the best Maximin option.

Questions/Tasks



C9 Apply the filtering rules using the supplied results. Construct a step-by-step filtering table showing the rule, relevant numerical values, and alternatives retained and eliminated at each stage, including the final decision.

Solution

# Step	Description	Left Alternatives	Eliminated Alternatives
0	Initial alternatives	$\{X_1, X_2, X_3, X_4, X_5, X_6\}$	—
1	EV: keep the 3 highest values.	$X_4(91.25)$, $X_2(85.25)$, $X_5(84.75)$	$X_1(80.50)$, $X_3(77.75)$, $X_6(84.15)$
2	Minimax Regret: keep the 2 smallest remaining maximum regrets. Retain both tied values of 20.	$X_2(20)$, $X_5(20)$	$X_4(30)$
3	Maximin: select the highest remaining value.	**Selected:** $X_5(45)$	$X_2(40)$



2.3 Trade Finance Guarantees

Problem Description

A trade finance agency chooses one guarantee policy among $G_1, G_2, G_3, G_4, G_5, G_6$. States and probabilities are shown below. Payoffs are in thousand USD. Policies G_4 and G_5 balance upside and downside because they diversify by sector, while G_3 is highly cyclical.

State	Probability	G_1	G_2	G_3	G_4	G_5	G_6
M_1	0.45	100	94	114	90	104	96
M_2	0.35	72	78	64	84	70	80
M_3	0.20	32	44	12	56	38	26

The summary statistics below are already calculated. Use them directly; do not recompute them from the payoffs.

Alternative	EV	Maximin	Maximum Regret
G_1	78.10	32	24
G_2	81.20	44	20
G_3	79.70	12	44
G_4	83.90	56	24
G_5	81.70	38	20
G_6	75.40	26	34

Filtering rule: (1) keep best 4 EV options; (2) from those, keep best 2 Maximin options; (3) from those, choose smallest Minimax Regret.

Questions/Tasks



C9 Apply the filtering rules using the supplied results. Construct a step-by-step filtering table showing the rule, relevant numerical values, and alternatives retained and eliminated at each stage, including the final decision.

Solution

# Step	Description	Left Alternatives	Eliminated Alternatives
0	Initial alternatives	$\{G_1, G_2, G_3, G_4, G_5, G_6\}$	—
1	EV: keep the 4 highest values.	$G_4(83.90)$, $G_5(81.70)$, $G_2(81.20)$, $G_3(79.70)$	$G_1(78.10)$, $G_6(75.40)$
2	Maximin: keep the 2 highest remaining values.	$G_4(56)$, $G_2(44)$	$G_5(38)$, $G_3(12)$
3	Minimax Regret: select the smallest remaining maximum regret.	**Selected:** $G_2(20)$	$G_4(24)$



2.4 Insurance Portfolio Mixes

Problem Description

An insurance broker considers six portfolio mixes Y_1 to Y_6 under four climate-risk states with probabilities W_1:0.30, W_2:0.25, W_3:0.25, W_4:0.20. Payoffs are in thousand USD. Y_4 is conservative and protects downside in severe-weather years, while Y_3 has the highest upside but a weak stress outcome.

State	Probability	Y_1	Y_2	Y_3	Y_4	Y_5	Y_6
W_1	0.30	136	126	146	116	130	121
W_2	0.25	94	98	84	103	90	95
W_3	0.25	66	74	58	88	69	77
W_4	0.20	28	46	14	64	48	38

The summary statistics below are already calculated. Use them directly; do not recompute them from the payoffs.

Alternative	EV	Maximin	Maximum Regret
Y_1	84.90	28	40
Y_2	89.20	46	20
Y_3	82.20	14	50
Y_4	95.25	64	30
Y_5	88.70	48	20
Y_6	87.95	38	26

Filtering rule: (1) keep the best 3 EV options; (2) from those, keep the best 2 Minimax Regret options; (3) from those, choose the best Maximin option.

Questions/Tasks



C9 Apply the filtering rules using the supplied results. Construct a step-by-step filtering table showing the rule, relevant numerical values, and alternatives retained and eliminated at each stage, including the final decision.

Solution

# Step	Description	Left Alternatives	Eliminated Alternatives
0	Initial alternatives	$\{Y_1, Y_2, Y_3, Y_4, Y_5, Y_6\}$	—
1	EV: keep the 3 highest values.	$Y_4(95.25)$, $Y_2(89.20)$, $Y_5(88.70)$	$Y_1(84.90)$, $Y_3(82.20)$, $Y_6(87.95)$
2	Minimax Regret: keep the 2 smallest remaining maximum regrets. Retain both tied values of 20.	$Y_2(20)$, $Y_5(20)$	$Y_4(30)$
3	Maximin: select the highest remaining value.	**Selected:** $Y_5(48)$	$Y_2(46)$

